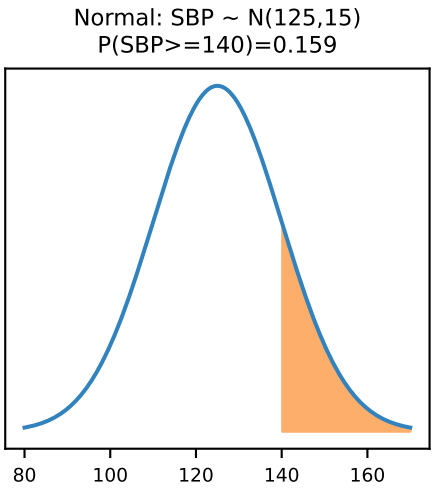
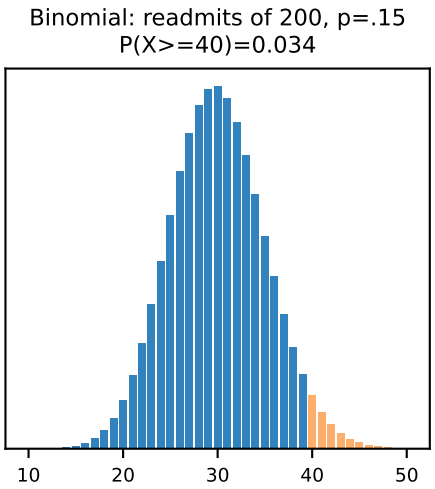
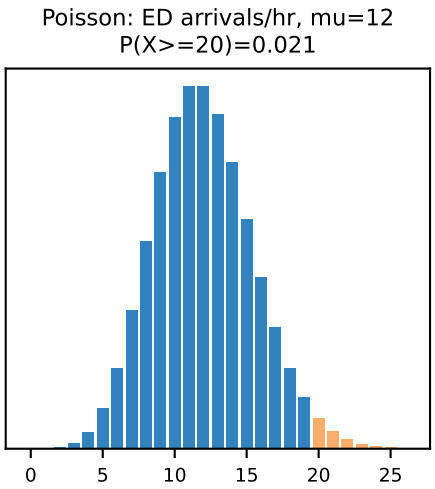
Day 2 Cheat Sheet: Distributions & Hypothesis Tests

Is the PPO vs. HMO cost gap real, or just noise?







THREE DISTRIBUTIONS YOU WILL SEE IN HEALTHCARE
- Normal: continuous, symmetric (lab values, blood pressure)
- Binomial: count of yes/no events out of n (readmissions)
- Poisson: count of events per time window (ED arrivals)
- .pdf/.pmf = chance of one value; .cdf = P(X <= x); .sf = P(X > x)

HYPOTHESIS TEST IN 5 STEPS
1. H0 (null): no difference. H1: there is a difference.
2. Pick alpha before looking (usually 0.05).
3. Compute a test statistic (Welch t, or a permutation gap).
4. p-value = P(a result this extreme | H0 is true).
5. p < alpha: reject H0. p >= alpha: fail to reject (NOT "proven equal").

A p-value is NOT the chance H0 is true, and NOT the effect size.
Central Limit Theorem: means of samples look normal even when data is skewed.

THIS DATA (n=5,000 simulated claims)
HMO mean $679, median $408 | PPO mean $743, median $429
Welch t-test on log1p(amount): t = 2.08, p = 0.037 -> reject H0 at 0.05
Permutation test on medians: gap $20, p = 0.097 -> fail to reject

```
from scipy import stats
stats.norm.sf(140, loc=125, scale=15)      # P(X >= 140)
stats.binom.sf(39, n=200, p=0.15)          # P(X >= 40)
stats.poisson.sf(19, mu=12)                # P(X >= 20)
t, p = stats.ttest_ind(ppo, hmo, equal_var=False)  # Welch
s = rng.permutation(labels)  # shuffle labels -> null world
```

Infographic: How a Hypothesis Test Decides

Simulated healthcare claims, PPO vs. HMO

H0
No cost difference

alpha
0.05 (set first)

Test
Welch t on log cost

p-value
0.037

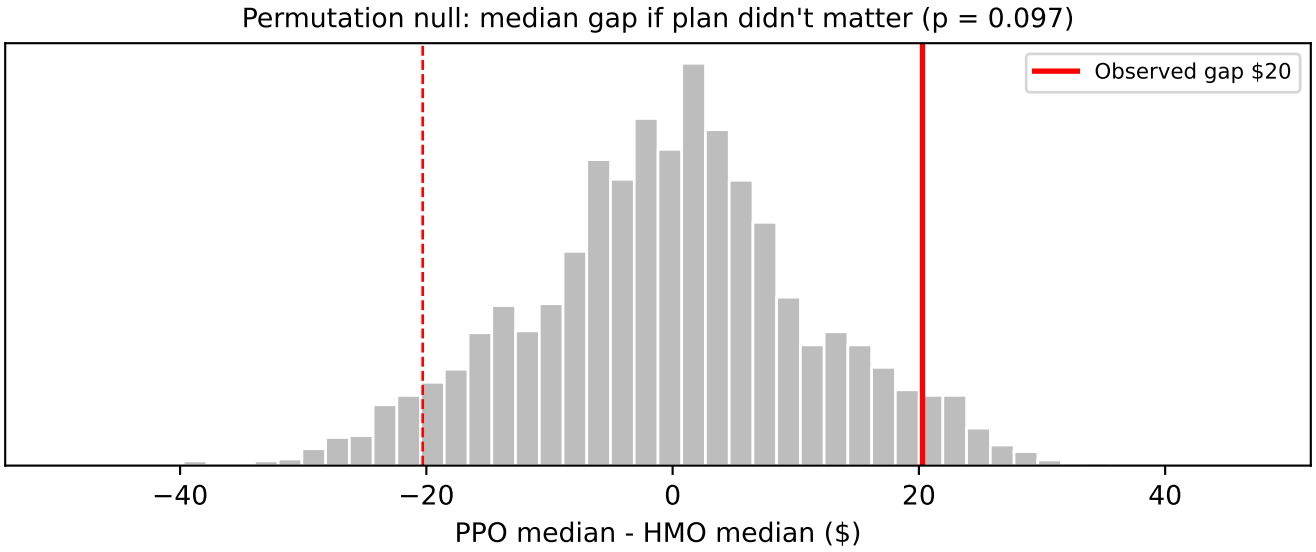
Decision
p < 0.05 reject H0

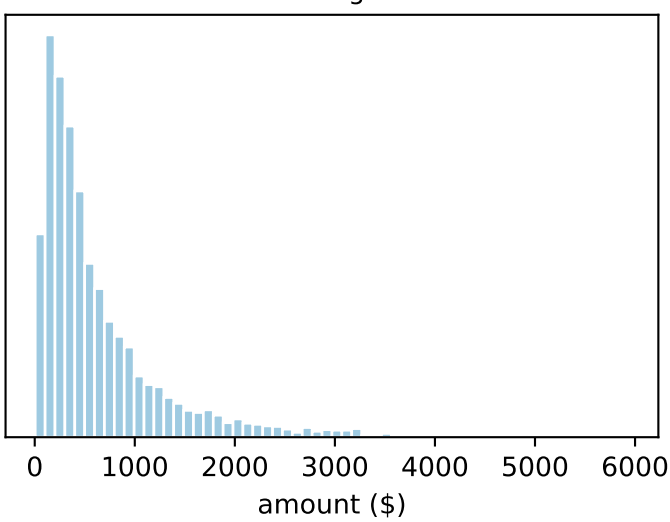
MEAN GAP
$64

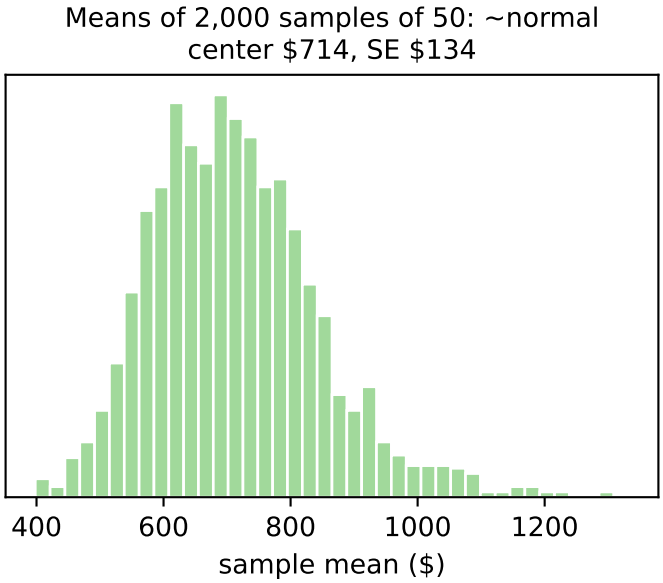
WELCH p
0.037

MEDIAN GAP
$20

PERM p
0.097







Two tests, two questions: the log-mean gap clears 0.05, the median gap does not. Report the effect size, not just p.

Hands-On Worksheet: ds-genai-90-days / day02

60-90 minutes | notebook: day02/distributions_hypothesis.ipynb

[ ] mkdir day02, open VS Code, conda activate ds90 (scipy is in the Day 1 env)

[ ] Normal / Binomial / Poisson: compute the 3 tail probabilities with scipy.stats

[ ] Plot each distribution and shade the tail you computed

[ ] Simulate 5,000 claims (seed 42, PPO costs x1.10), add log_amt = np.log1p(amount)

[ ] Write H0, H1 and alpha = 0.05 in a markdown cell BEFORE testing

[ ] Welch t-test on log_amt: stats.ttest_ind(ppo, hmo, equal_var=False)

[ ] Permutation test on the median gap (5,000 shuffles), plot the null histogram

[ ] CLT: 2,000 sample means (n=50), plot next to the raw claims histogram

[ ] Write 3 stakeholder insights: effect size + p-value + caveat

[ ] git add day02 && git commit -m 'Day 2: distributions and hypothesis tests' && git push

[ ] Stretch: Snowflake: AVG / STDDEV / COUNT of LN(1+amount) by plan, compute t by hand

```
import numpy as np, pandas as pd
from scipy import stats
rng = np.random.default_rng(42); n = 5000
plan = rng.choice(["HMO", "PPO"], n)
amount = rng.lognormal(6, 1, n) * np.where(plan == "PPO", 1.10, 1.0)
df = pd.DataFrame({"plan_type": plan, "amount": amount})
df["log_amt"] = np.log1p(df["amount"])
hmo = df.loc[df.plan_type == "HMO", "log_amt"]
ppo = df.loc[df.plan_type == "PPO", "log_amt"]
t, p = stats.ttest_ind(ppo, hmo, equal_var=False)
print(f"t = {t:.2f}, p = {p:.3f}")   # expect t = 2.08, p = 0.037
```

My H0: _____ alpha: _____

My p-values: Welch _____ Permutation _____ Decision: _____

Day 2 Self-Check Quiz

1. Daily count of ED arrivals: Normal, Binomial, or Poisson? Why?
   _____

2. What does stats.binom.sf(39, n=200, p=0.15) return in plain words?
   _____

3. p = 0.037. Does that mean a 3.7% chance H0 is true? What does it mean?
   _____

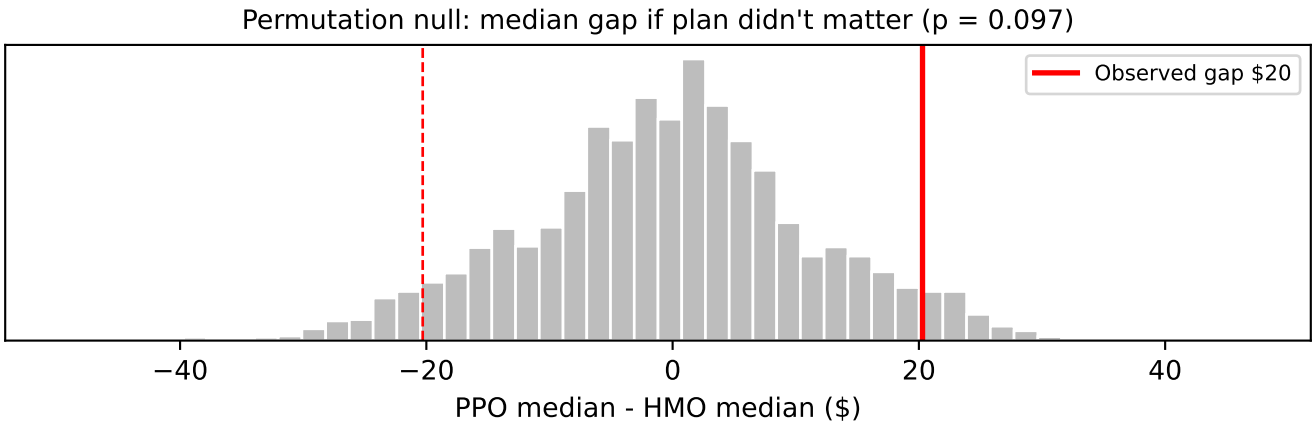
4. p = 0.20 at alpha 0.05. Have you proven HMO and PPO cost the same?
   _____

5. Why use Welch's t-test (equal_var=False) instead of the classic t-test?
   _____

6. What does shuffling plan labels in a permutation test simulate?
   _____

7. Raw claims are skewed. Why do sample means of 50 claims look normal?
   _____

8. With 5 million claims, a $2 gap is 'significant'. Should you act on it?
   _____



Answers: 1 Poisson (counts per time window) 2 P(40 or more readmissions of 200) = 0.034
3 No: it is P(data this extreme | H0 true) 4 No: you only failed to reject H0 5 does not assume equal variances
6 a world where plan type has no effect (the null) 7 Central Limit Theorem 8 not by itself: check effect size and cost